

Minesweeper Solver

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Abstract: The project involves using machine learning and proper algorithms in order to complete a game of minesweeper

Introduction: I created a program that uses machine learning in order to complete a game of mine sweeper. It will scan the board and search for moves it can play. It can solve any minesweeper board.

Project Methods: I used a previously created minesweeper program for Python and modified it so it connects to a solver. In order to work, start the minesweeper program, start the solver and select a difficulty on the minesweeper program. The game sends its current visible status to the solver and the solver queues up and plays the determinate moves from that. Once all the moves have been played, the solver will again look for determinate moves. If it can not find any, it will play probabilistic moves, which can fail.

Conclusion: The program will work unless it is forced to make probabilistic moves. Then there is a chance it could fail

Code: <https://github.com/JoshuaKlot/CSProject>

Future plans: Rather than a difficulty selector, I would like the minesweeper program to have the user input the number of rows, columns and mines that the solver is aware of. I think this will help the solver make more informed decisions.

References/Resources:

MineSweeper Program: <https://github.com/Dest76ter/MineSweeper-using-Python-Tkinter>

Minesweeper algorithm:

<https://dash.harvard.edu/entities/publication/73120378-eb34-6bd4-e053-0100007fdf3b>

Programs used:

Python

Visual Studio Code